

$$\begin{aligned}
& -\frac{1}{168} \mathbf{g}_1^4 \mathbf{y}_d^{i112} \mathbf{L}_{F2,2,-1}[\mathbf{m}_1, \mathbf{m}_q^{i12}] + \frac{1}{18} \mathbf{g}_1^2 \mathbf{y}_d^{p12} \overline{\mathbf{y}}_0^{\text{pr}} \mathbf{y}_0^{i1r} \mathbf{L}_{F3,1,-1}[\mathbf{m}_1, \mathbf{m}_q^p] + \\
& \mathbf{a}_d^{p12} \overline{\mathbf{y}}_0^{\text{pr}} \mathbf{y}_0^{i1r} \mathbf{L}_{F4,1,-2}[\mathbf{m}_1, \mathbf{m}_q^p] + \frac{1}{864} \mathbf{g}_1^4 \mathbf{y}_d^{i112} (\mathbf{g}_1^2 - 3 \mathbf{g}_2^2) \mathbf{L}_{F2,2,-1}[\mathbf{m}_1, \mathbf{m}_q^{i11}] + \\
& \mathbf{a}_d^{p12} \overline{\mathbf{y}}_0^{\text{pr}} \mathbf{y}_0^{i1r} \mathbf{L}_{F2,1,0}[\mathbf{m}_1, \mathbf{m}_q^p] - \frac{8}{9} \mathbf{g}_1^2 \mathbf{y}_d^{p12} \overline{\mathbf{y}}_0^{\text{pr}} \mathbf{y}_0^{i1r} \mathbf{L}_{F3,1,-1}[\mathbf{m}_1, \mathbf{m}_q^p] + \\
& \mathbf{a}_d^{p12} \overline{\mathbf{y}}_0^{\text{pr}} \mathbf{y}_0^{i1r} \mathbf{L}_{F4,1,-2}[\mathbf{m}_1, \mathbf{m}_q^p] + \frac{3}{8} \mathbf{g}_2^2 \mathbf{y}_d^{p12} \overline{\mathbf{y}}_0^{\text{pr}} \mathbf{y}_0^{i1r} \mathbf{L}_{F2,1,0}[\mathbf{m}_2, \mathbf{m}_q^p] - \\
& \mathbf{a}_d^{p12} \overline{\mathbf{y}}_0^{\text{pr}} \mathbf{y}_0^{i1r} \mathbf{L}_{F3,1,-1}[\mathbf{m}_2, \mathbf{m}_q^p] + \frac{1}{2} \mathbf{g}_1^2 \mathbf{y}_d^{p12} \overline{\mathbf{y}}_0^{\text{pr}} \mathbf{y}_0^{i1r} \mathbf{L}_{F4,1,-2}[\mathbf{m}_2, \mathbf{m}_q^p] + \\
& \mathbf{a}_d^{i112} (\mathbf{g}_1^2 + \mathbf{g}_2^2) \mathbf{L}_{F2,2,-1}[\mathbf{m}_2, \mathbf{m}_q^{i12}] + \frac{1}{9} \mathbf{g}_1^2 \mathbf{g}_2^2 \mathbf{y}_d^{i112} \mathbf{L}_{F2,2,-1}[\mathbf{m}_3, \mathbf{m}_q^{i12}] + \\
& \mathbf{a}_d^{p12} \overline{\mathbf{y}}_0^{\text{pr}} \mathbf{y}_0^{i1r} \mathbf{L}_{F2,1,0}[\mathbf{m}_3, \mathbf{m}_q^p] - \frac{8}{9} \mathbf{g}_3^2 \mathbf{y}_d^{p12} \overline{\mathbf{y}}_0^{\text{pr}} \mathbf{y}_0^{i1r} \mathbf{L}_{F3,1,-1}[\mathbf{m}_3, \mathbf{m}_q^p] + \\
& \mathbf{a}_d^{p12} \overline{\mathbf{y}}_0^{\text{pr}} \mathbf{y}_0^{i1r} \mathbf{L}_{F4,1,-2}[\mathbf{m}_3, \mathbf{m}_q^p] + \frac{1}{18} \mathbf{g}_2^2 \mathbf{y}_d^{i112} (\mathbf{g}_1^2 - 3 \mathbf{g}_2^2) \mathbf{L}_{F2,2,-1}[\mathbf{m}_3, \mathbf{m}_q^{i11}] + \\
& \mathbf{a}_d^{p12} \overline{\mathbf{y}}_0^{\text{pr}} \mathbf{y}_0^{i1r} \mathbf{L}_{F2,1,0}[\mathbf{m}_3, \mathbf{m}_q^p] - \frac{8}{9} \mathbf{g}_3^2 \mathbf{y}_d^{p12} \overline{\mathbf{y}}_0^{\text{pr}} \mathbf{y}_0^{i1r} \mathbf{L}_{F3,1,-1}[\mathbf{m}_3, \mathbf{m}_q^p] + \\
& \mathbf{a}_d^{p12} \overline{\mathbf{y}}_0^{\text{pr}} \mathbf{y}_0^{i1r} \mathbf{L}_{F4,1,-2}[\mathbf{m}_3, \mathbf{m}_q^p] + \frac{1}{2} \mathbf{g}_1^2 \mathbf{y}_d^{i112} \overline{\mathbf{a}}_0^{\text{pr}} \mathbf{a}_d^{\text{pr}} \mathbf{L}_{F2,2,0}[\mathbf{m}_d^r, \mathbf{m}_q^p] - \\
& \overline{\mathbf{a}}_0^{\text{pr}} \mathbf{y}_d^{i12} \mathbf{y}_d^{i1r} \mathbf{L}_{F2,1,0}[\mathbf{m}_d^r, \tilde{\mu}] + \frac{1}{24} \mathbf{g}_1^2 \overline{\mathbf{y}}_0^{\text{pr}} \mathbf{y}_d^{p12} \mathbf{y}_d^{i1r} \mathbf{L}_{F2,2,-1}[\mathbf{m}_d^r, \tilde{\mu}] - \\
& \mathbf{a}_d^{i112} \mathbf{L}_{F2,1,0}[\mathbf{m}_2^r, \mathbf{m}_1] - \frac{2}{3} \mathbf{g}_1^2 \mathbf{g}_3^2 \mathbf{y}_d^{i112} \mathbf{L}_{F2,1,0}[\mathbf{m}_d^{i12}, \mathbf{m}_3] + \\
& \mathbf{a}_d^{i112} \overline{\mathbf{a}}_e^{\text{pr}} \mathbf{a}_e^{\text{pr}} \mathbf{L}_{F2,2,0}[\mathbf{m}_e^r, \mathbf{m}_1^p] - \frac{1}{2} \mathbf{g}_2^2 \mathbf{y}_d^{i112} \overline{\mathbf{a}}_e^{\text{pr}} \mathbf{a}_e^{\text{pr}} \mathbf{L}_{F3,1,0}[\mathbf{m}_e^r, \mathbf{m}_e^r] - \\
& \mathbf{a}_d^{i112} \overline{\mathbf{a}}_e^{\text{pr}} \mathbf{a}_e^{\text{pr}} \mathbf{L}_{F3,2,-1}[\mathbf{m}_e^r, \mathbf{m}_e^r] - \frac{8}{9} \mathbf{y}_d^{i112} \overline{\mathbf{a}}_e^{\text{pr}} \mathbf{a}_e^{\text{pr}} (\mathbf{g}_1^2 - 3 \mathbf{g}_2^2) \mathbf{L}_{F4,1,-1}[\mathbf{m}_e^r, \mathbf{m}_e^r] + \\
& \overline{\mathbf{a}}_e^{\text{pr}} \mathbf{a}_e^{\text{pr}} (\mathbf{g}_1^2 - \mathbf{g}_2^2) \mathbf{L}_{F5,1,-2}[\mathbf{m}_e^r, \mathbf{m}_e^r] + \\
& \overline{\mathbf{a}}_d^{\text{pr}} \mathbf{a}_d^{\text{pr}} (2 \mathbf{g}_1^2 - 3 \mathbf{g}_2^2) \mathbf{L}_{F3,1,0}[\mathbf{m}_q^p, \mathbf{m}_d^r] - \frac{1}{4} \mathbf{g}_1^2 \mathbf{y}_d^{i112} \overline{\mathbf{a}}_d^{\text{pr}} \mathbf{a}_d^{\text{pr}} \mathbf{L}_{F3,2,-1}[\mathbf{m}_q^p, \mathbf{m}_d^r] - \\
& \overline{\mathbf{a}}_d^{\text{pr}} \mathbf{a}_d^{\text{pr}} (7 \mathbf{g}_1^2 - 9 \mathbf{g}_2^2) \mathbf{L}_{F4,1,-1}[\mathbf{m}_q^p, \mathbf{m}_d^r] + \\
& \overline{\mathbf{a}}_d^{\text{pr}} \mathbf{a}_d^{\text{pr}} (\mathbf{g}_1^2 - \mathbf{g}_2^2) \mathbf{L}_{F5,1,-2}[\mathbf{m}_q^p, \mathbf{m}_d^r] + \\
& \mathbf{a}_d^{i112} \overline{\mathbf{y}}_0^{\text{pr}} \mathbf{y}_0^{\text{pr}} (\mathbf{g}_1^2 + 3 \mathbf{g}_2^2) \mathbf{L}_{F2,2,0}[\mathbf{m}_q^p, \mathbf{m}_d^r] - \frac{1}{12} \mathbf{g}_1^2 \overline{\mathbf{y}}_0^{\text{pr}} \mathbf{y}_d^{p12} \mathbf{y}_d^{i1r} \mathbf{L}_{F2,1,0}[\mathbf{m}_q^p, \tilde{\mu}] + \\
& \overline{\mathbf{a}}_d^{\text{pr}} \mathbf{y}_d^{i12} \mathbf{y}_d^{i1r} \mathbf{L}_{F2,2,-1}[\mathbf{m}_q^p, \tilde{\mu}] - \frac{1}{432} \mathbf{g}_1^2 \mathbf{y}_d^{i112} (\mathbf{g}_1^2 - 3 \mathbf{g}_2^2) \mathbf{L}_{F2,1,0}[\mathbf{m}_1^r, \mathbf{m}_1] - \\
& \mathbf{a}_d^{i112} (\mathbf{g}_1^2 + \mathbf{g}_2^2) \mathbf{L}_{F2,1,0}[\mathbf{m}_2^r, \mathbf{m}_2] - \frac{1}{9} \mathbf{g}_3^2 \mathbf{y}_d^{i112} (\mathbf{g}_1^2 - 3 \mathbf{g}_2^2) \mathbf{L}_{F2,1,0}[\mathbf{m}_q^{i11}, \mathbf{m}_3] - \\
& \mathbf{a}_d^{i112} \overline{\mathbf{y}}_0^{\text{pr}} \mathbf{y}_0^{\text{pr}} (\mathbf{g}_1^2 + 3 \mathbf{g}_2^2) \mathbf{L}_{F3,1,0}[\mathbf{m}_d^r, \mathbf{m}_q^p] - \frac{1}{8} \mathbf{g}_2^2 \mathbf{y}_d^{i112} \mathbf{g}_2^{\text{pr}} \mathbf{y}_d^{i1r} \mathbf{L}_{F2,2,-1}[\mathbf{m}_3, \mathbf{g}_2^2] \\
& - [\mathbf{m}_1^r, \mathbf{m}_q^p] - \frac{3}{2} \tilde{\mu}^2 \mathbf{y}_d^{i112} \overline{\mathbf{y}}_0^{\text{pr}} \mathbf{y}_0^{\text{pr}} (\mathbf{g}_1^2 - 3 \mathbf{g}_2^2) \mathbf{L}_{F4,1,-1}[\mathbf{m}_0^r, \mathbf{m}_q^p] + \\
& \mathbf{a}_d^{i112} \overline{\mathbf{y}}_0^{\text{pr}} \mathbf{y}_0^{\text{pr}} (\mathbf{g}_1^2 - \mathbf{g}_2^2) \mathbf{L}_{F5,1,-2}[\mathbf{m}_0^r, \mathbf{m}_q^p] + \frac{1}{6} \mathbf{g}_1^2 \mathbf{y}_d^{p12} \overline{\mathbf{y}}_0^{\text{pr}} \mathbf{y}_0^{i1r} \mathbf{L}_{F2,1,0}[\mathbf{m}_0^r, \tilde{\mu}] - \\
& \overline{\mathbf{a}}_d^{\text{pr}} \mathbf{y}_d^{i12} \overline{\mathbf{y}}_0^{\text{pr}} \mathbf{y}_0^{i1r} \mathbf{L}_{F2,2,-1}[\mathbf{m}_0^r, \tilde{\mu}] - \frac{3}{2} \overline{\mathbf{y}}_0^{\text{pr}} \mathbf{y}_d^{\text{sr}} \mathbf{y}_d^{i112} \overline{\mathbf{y}}_0^{\text{st}} \mathbf{y}_0^{\text{pt}} \mathbf{L}_{F2,1,0}[\mathbf{m}_0^{\text{t}}, \mathbf{m}_d^r] - \\
& \overline{\mathbf{y}}_d^{\text{sr}} \mathbf{y}_d^{i112} \overline{\mathbf{y}}_0^{\text{st}} \mathbf{y}_0^{\text{pt}} \mathbf{L}_{F3,1,-1}[\mathbf{m}_0^{\text{t}}, \mathbf{m}_d^r] + \frac{1}{2} \mathbf{g}_1^2 \mathbf{y}_d^{i1$$